

BOMB DISPOSAL ROBOT

A

Minor Project-I Report

Submitted in partial fulfilment of the requirement for the award of
Degree of

Bachelor of Technology in
ELECTRONICS & COMMUNICATION ENGINEERING

Submitted to



RAJIVGANDHIPROUDYOGIKI VISHWAVIDHYALAYA, BHOPAL
(M.P.)

Submitted by

Dharmendra Belwanshi (0537EC211021)

Nikhil Khatarkar (0537EC211034)

Amit Athiya(0537EC211006)

Rushali Sonekar (0537CE211062)

Ankit Hanotre (0537EC211008)

Raj Kumar Saw(0537EC213D01)

Under the Supervision of

Prof. Ritesh Dhiman



Department of Electronics & Communication Engineering

Sagar Institute of Science, Technology & Research (SISTec-R)

Bhopal
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Sagar Institute of Science, Technology & Research (SISTec-R), Bhopal

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING



CERTIFICATE

This is to certify that the work embodies in this project entitled “Bomb Disposal Robot” being submitted by Dharmendra Belwanshi (0537EC211021), Nikhil Khatarkar (0537EC2110034), Amit Athiya (0537EC211006), Ankit Hanotre (0537EC211008) , Raj Kumar Saw (0537EC213D01) and Rushali Sonekar (0537CE211062) in partial fulfilment of the requirement for the award of degree of B.Tech. (Electronics & Communication Engineering), to Rajiv Gandhi Proudhyogiki Vishwavidyalaya is a record of a Bonafide piece of work, carried out by them under my supervision and guidance.

Guided & Approved by:

Prof. Ritesh Dhiman
Assistant Professor
Electronics & Communication Engineering

Forwarded by:-

Prof. Harvinder Singh Saluja
HOD-ECE
SISTec-R, Bhopal

Dr. Jyoti Deshmukh
Principal
SISTec-R, Bhopal

Sagar Institute of Science, Technology & Research (SISTec-R), Bhopal



DECLARATION

We, Dharmendra Belwanshi (0537EC211021), Nikhil Khatarkar (0537EC2110034), Amit Athiya (0537EC211006) Ankit Hanotre (0537EC211008) , Raj Kumar Saw (0537EC213D01) and Rushali Sonekar (0537CE211062) students of B.Tech. in Electronics & Communication Engineering, session 2021-25, Sagar Institute of Science, Technology & Research, Bhopal (M.P.), hereby declare that the work presented in this project entitled “Bomb Disposal Robot” is the outcome of our own work, is Bonafide and correct to the best of our knowledge and this work has been carried out taking care of Engineering Ethics. The work presented does not infringe any patented work and has not been submitted to any other University or anywhere else for the award of any degree or any professional diploma.

Dharmendra Belwanshi(0537EC211021)

Nikhil Khatarkar(0537EC211034)

Amit Athiya(0537EC211006)

Rushali Sonekar(0537EC211062)

Ankit Hanotre (0537EC211008)

Raj Kumar Saw(0537EC213D01)

Date:

Sagar Institute of Science, Technology & Research (SISTec-R), Bhopal

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING



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Dharmendra Belwanshi(0537EC211021)

Nikhil Khatarkar(0537EC211034)

Amit Athiya(0537EC211006)

Rushali Sonekar(0537EC211062)

Ankit Hanotre(0537EC211008)

Raj kumar Saw(0537EC213D01)

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1. ABSTRACT:-

The advent of technological advancements has led to the development of innovative solutions for hazardous tasks, including bomb disposal. This project aims to design and implement a remotely operated robot capable of safely navigating treacherous environments and executing precise tasks, particularly those involving explosive devices.

The robot's mechanical design incorporates a robust chassis equipped with tracked locomotion for optimal manoeuvrability on various terrains. A versatile robotic arm, equipped with grippers and cutting tools, is integrated to enable the manipulation and neutralization of explosive devices. The integration of a camera and sensor suite, including infrared and ultrasonic sensors, provides real-time visual and sensory feedback to the operator.

The control system, implemented using a microcontroller, facilitates wireless remote control, allowing the operator to manoeuvre the robot and control the robotic arm from a safe distance. The system incorporates advanced algorithms for obstacle avoidance, path planning, and precise control of the robotic arm's movements.

A comprehensive sensor suite is integrated into the robot to provide real-time information about the robot's surroundings and the target device. High-resolution cameras enable visual inspection of the scene and the device. Infrared sensors allow for detection of heat signatures, which can be indicative of certain types of explosive devices. Laser rangefinders provide accurate distance measurements, aiding in navigation and precise manipulation tasks. Additionally, chemical and biological sensors can be integrated to detect hazardous substances associated with explosive devices.

The project emphasizes safety, reliability, and user-friendliness. Rigorous testing and validation procedures are employed to ensure the robot's performance and adherence to safety standards. By developing and deploying such a robot, we aim to mitigate risks to human life and property in bomb disposal operations, enhancing public safety and security.

The increasing prevalence of explosive threats poses a significant risk to public safety. To address this challenge, the development of advanced robotic systems capable of safely and efficiently handling explosive devices is imperative. This project aims to design, develop, and implement a remotely operated bomb disposal robot that can navigate hazardous environments, detect and identify explosive threats, and neutralize them in a controlled manner.

2. INTRODUCTION:-

The Bomb Disposal Robot Project is aimed at developing a reliable, efficient, and robust robotic system capable of identifying, handling, and neutralizing explosive devices. The robot incorporates advanced features such as remote operation, real time video surveillance, robotic arm manipulation, and precision control to safely neutralize threats in a controlled manner.

In today's world, the increasing threat of terrorism and explosive hazards has necessitated the development of advanced tools and technologies to ensure public safety. Bomb disposal robots play a critical role in mitigating risks to human life while handling explosive devices. These robotic systems are designed to perform intricate and high-stakes tasks in hazardous environments, reducing the need for human intervention in potentially life-threatening situations.

At its core, the robot integrates multiple subsystems to ensure functionality and versatility. The mechanical framework is built to sustain rough terrains and provide stability during operations. The robotic arm, equipped with a claw or gripper, facilitates safe handling of suspicious objects. A camera module coupled with wireless transmission provides the operator with real-time visuals, enabling precise decision-making. The integration of sensors, such as metal detectors and temperature sensors, enhances the robot's ability to detect explosives and assess environmental risks.

This project focuses on developing a scalable and cost-effective solution to address the challenges faced by bomb disposal teams globally. It aligns with the objective of safeguarding human lives while improving efficiency in handling explosive threats. Through this initiative, the project aims to push the boundaries of robotics, integrating modern technology with innovative design to create a reliable partner in hazardous operations.

In addition to technical aspects, this project emphasizes ethical considerations, ensuring compliance with safety standards and operational guidelines. By simulating real-world scenarios and conducting extensive testing, the robot is designed to meet the practical demands of bomb disposal operations.

The development of the bomb disposal robot not only addresses immediate security concerns but also contributes to advancements in robotics and automation. This innovation underscores the role of technology in enhancing safety and efficiency in high-risk environments, offering a solution that is both functional and lifesaving. Through interdisciplinary collaboration and research, this project aims to pave the way for future innovations in robotic systems designed for public safety and security.

3. PRINCIPLE IDEA: -

The principle idea of the Bomb Disposal Robot Project is to develop a robotic system that ensures the safe disposal of explosive devices while minimizing risks to human life. This is achieved through the integration of advanced robotics, sensor technology, and remote-control capabilities. The robot operates as a remote extension of a human operator, designed to function efficiently in hazardous environments.

The robot's mobility system enables it to traverse various terrains, including stairs, rough surfaces, and debris. This ensures accessibility to locations where explosives are typically placed. A robust robotic arm equipped with precise manipulators allows the robot to handle, disarm, or relocate explosive devices with accuracy.

To enhance situational awareness, the robot is equipped with a high-resolution camera system providing real-time video feedback to the operator. Additional sensors, such as metal detectors and gas sensors, help detect and analyse the presence of explosives.

Remote operation is facilitated through wireless communication, allowing operators to control the robot from a safe distance. The system may include semi-autonomous features for navigation and obstacle avoidance, ensuring efficiency in dynamic environments.

Safety mechanisms, including fail-safes and redundant communication systems, are integrated to maintain reliability during critical missions. The robot's design prioritizes durability, precision, and adaptability, making it suitable for both civilian and military applications.

This project aims to replace traditional, high-risk human bomb disposal operations with a safer, technology-driven alternative, highlighting the potential of robotics to enhance public safety.

4. Block diagram: -

A bomb disposal robot is a remotely operated device designed to handle and neutralize explosives safely. It consists of a robust chassis for mobility, a manipulator arm with a gripper, and sensors like cameras and proximity detectors for situational awareness. Controlled wirelessly, the operator guides the robot to the bomb site. Live video and sensor feedback enable precise navigation and object handling. The manipulator arm is used to pick up or neutralize the explosive using specialized tools. Powered by batteries, the robot ensures safe operation in hazardous environments, reducing human risk during bomb disposal missions.

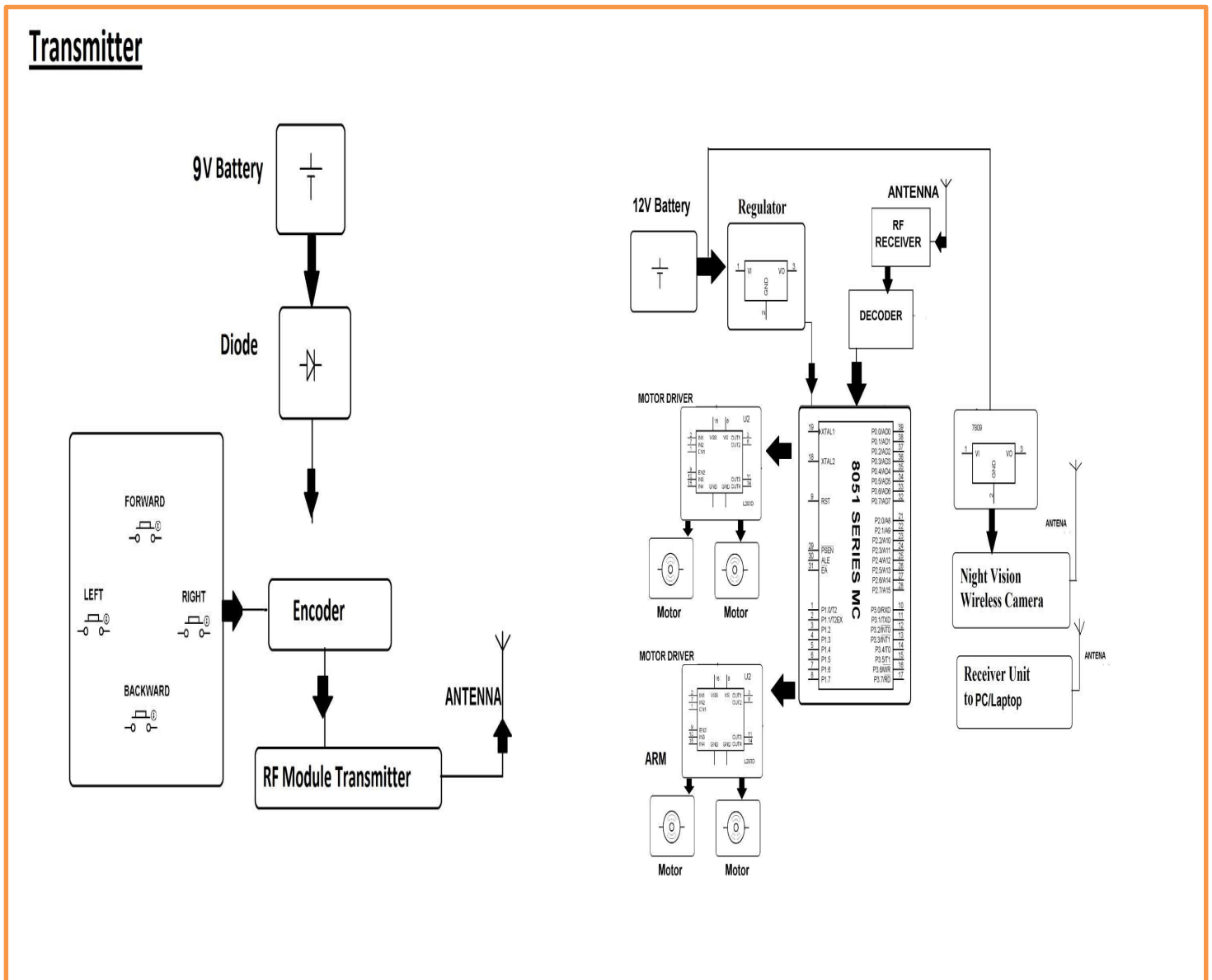


Fig 4.1 - Block Diagram

5. Details of components: -

The major components are: -

1. Arduino Uno Board
2. L298N Motor Driver
3. Gripper (Robotic arm)
4. RF Controller Tx Rx
5. LM2596 DC-DC Buck Converter
6. Switch
7. Jumper Wire
8. Wheel Screws
9. Nuts &Nuts
10. Motor
11. Robot Body Case
12. Servo Motors

5.1 Arduino Uno:

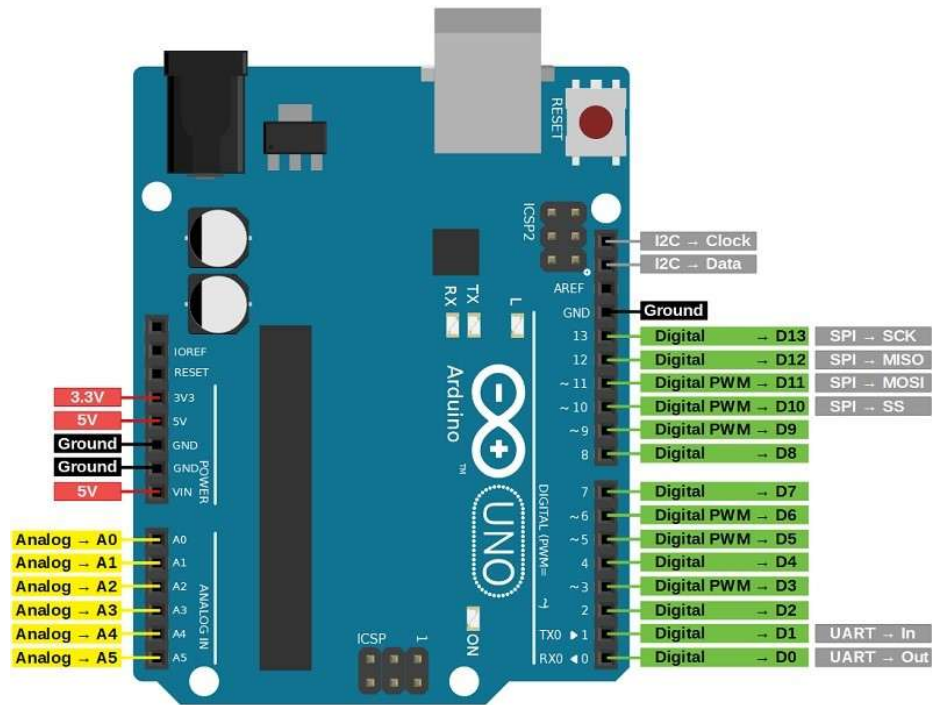


Fig 5.1 - Arduino Uno

The Arduino Uno shown in Fig 5.1, microcontroller board was utilized as the central component due to its versatility and ease of integration. The Arduino Uno, based on the ATmega328P microcontroller, offers 14 digital input/output pins and 6 analog input pins, making it highly suitable for controlling a variety of sensors, actuators, and other peripherals. The board operates at 5V and can be powered through a USB connection or an external power source, ensuring flexibility in different environments. It features built-in LEDs for easy debugging and a reset button for quick reboots during testing. The Arduino Uno was programmed using the Arduino IDE, which supports C/C++ programming languages and provides a vast library of resources to streamline development. Its reliable performance, combined with strong community support,

made the Arduino Uno an ideal choice for this project, enabling efficient prototyping and implementation of the desired functionalities.

Vin: This is the input voltage pin of the Arduino board used to provide input supply from an external power source.

5V: This pin of the Arduino board is used as a regulated power supply voltage and it is used to give supply to the board as well as onboard components.

3.3V: This pin of the board is used to provide a supply of 3.3V which is generated from a voltage regulator on the board

GND: This pin of the board is used to ground the Arduino board.

Reset: This pin of the board is used to reset the microcontroller. It is used to Resets the microcontroller.

Analog Pins: The pins A0 to A5 are used as an analog input and it is in the range of 0-5V.

Digital Pins: The pins 0 to 13 are used as a digital input or output for the Arduino board.

5.1.1 Application of Arduino Uno:

1. Home Automation: Controls lights, fans, and security systems in smart home projects.
2. Robotics: Operates motors, sensors, and communication modules in robots.
3. IoT Projects: Connects to the internet for remote monitoring and smart devices.
4. Education: Serves as a hands-on tool for learning electronics and programming.
5. Wearable Tech: Powers portable devices like fitness trackers and health monitors.

5.2 L298N Moter Driver :-

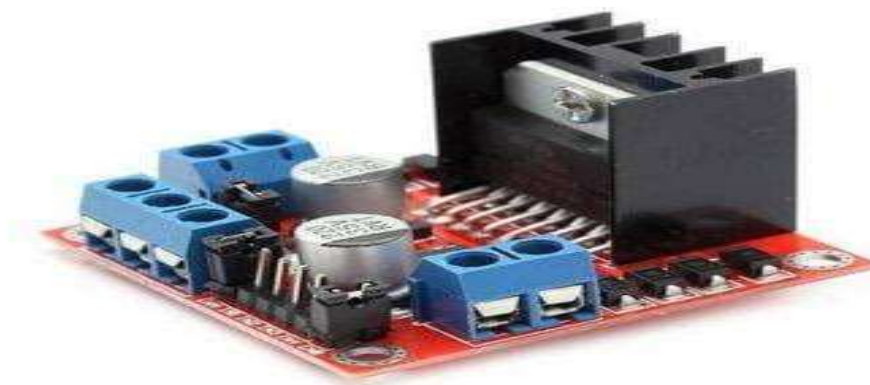


Fig 5.2 - L298N Moter Driver

A L298N Moter Driver in Fig 5.3, is a key component in the bomb disposal robot, enabling control of the robot's DC motors. It features a dual H-bridge design, allowing bidirectional control of two motors or one stepper motor.

Operating within a voltage range of 5V to 35V and supporting currents up to 2A per channel, it is suitable for most motor applications in the robot. The module integrates a 5V voltage regulator for powering other low-voltage components. It receives control signals from a microcontroller to regulate motor direction and speed. Direction is controlled via input pins (IN1, IN2 for Motor A; IN3, IN4 for Motor B), while speed is managed using PWM signals on the enable pins (ENA, ENB).

5.2.1 Advantages of L298N Motor Driver :

1. Compact and cost - effective.
2. Easy to integrate with standard microcontrollers.
3. High flexibility for multiple motor types.

5.3 Gripper (robotic arm):

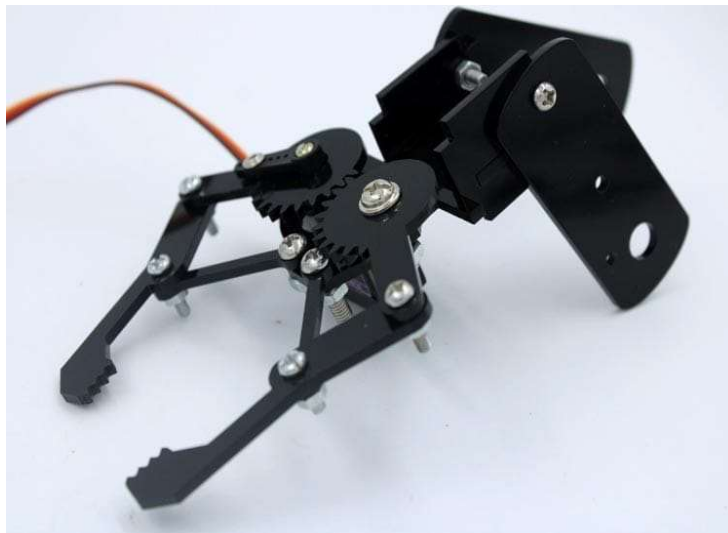


Fig 5.3 Gripper (robotic arm)

A Gripper (robotic arm) in Fig 5.3, is a critical component of a bomb disposal robot, designed for precise manipulation and handling of objects, including explosive devices, in hazardous environments. It typically consists of a robotic arm with multiple joints for flexibility and a gripper mechanism for securely holding objects. The arm is powered by motors (DC, servo, or stepper), with movements controlled by the robot's microcontroller. Sensors such as cameras and force sensors are integrated for accuracy and feedback. The gripper's primary function is to pick, hold, and neutralize bombs or suspicious items with minimal risk. The arm's design ensures it can reach confined spaces, providing versatility in bomb disposal scenarios.

The robotic arm is composed of multiple segments, usually connected with rotating or pivoting joints. These joints are powered by electric motors, such as DC motors or servo motors, which provide controlled movement. Depending on the design, the arm can have anywhere from three to six degrees of freedom (DOF), offering flexibility in reaching and positioning objects in various orientations.

5.3.1 Control and Operation:

The movements of the robotic arm and gripper are controlled by the robot's microcontroller or central processing unit (CPU). The control system receives input from the robot's operator and processes commands to control the arm's movement and the gripper's actions. Sensors, such as cameras or force sensors, provide feedback to the operator, enabling accurate positioning and force adjustment.

The arm can be controlled remotely via a wireless interface, typically using radio frequency (RF) or Wi-Fi signals. The operator views a live feed from cameras mounted on the robot and uses this feedback to guide the arm precisely to the target object. For bomb disposal, the gripper must be able to handle objects delicately to avoid triggering explosives while applying enough force to manipulate or disable a device.

5.4 RF Controller Tx Rx;



Fig 5.4 - RF Controller TX Rx

A RF Controller Tx Rx shown in Fig 5.4, is RF 433MHz is a pair of electronic RF transmitters and receiver modules used to send and receive between any two devices. The data is sent by the transmitter module from the transmitter end, and it receiver module at the receiver end. This RF signal will have a frequency of 433MHz. RF (Radio Frequency) communication plays a crucial role in enabling remote control of the robot. The system typically consists of two main parts: the transmitter (Tx) and the receiver (Rx). These components allow the operator to send control signals to the robot and receive feedback in real-time, ensuring safe and precise operation from a distance.

5.4.1 RF Transmitter (Tx):

The RF transmitter (Tx) is a handheld device or module that sends signals to the robot. It is equipped with a set of buttons, joysticks, or other input mechanisms that the operator uses to control the robot's movement and actions. The transmitter generates RF signals corresponding to the operator's inputs, which are then modulated and transmitted through the air to the robot's receiver. Common RF transmitter systems operate at frequencies such as 433 MHz or 2.4 GHz, depending on the range and application.

The transmitter typically includes:

1. **Control Inputs:** Buttons, switches, or joysticks that allow the user to control the robot's direction, speed, and gripper operation.
2. **Encoder/Modulator:** Encodes the operator's inputs into a modulated RF signal that can be transmitted effectively over a wireless medium.
3. **Power Supply:** Powers the transmitter, typically with a battery that provides long-duration operation for remote control.

5.4.2 RF Receiver (Rx):

The RF receiver (Rx) is installed on the robot and is responsible for receiving the RF signals from the transmitter. The receiver demodulates the incoming signals and translates them into control instructions that the robot's microcontroller can understand. These instructions direct the robot's actions, such as moving motors, activating the gripper, or steering the robot.

The receiver typically includes:

1. **Antenna:** Captures the incoming RF signals from the transmitter.

2. Demodulator: Decodes the received signals, converting them into digital or analog control signals.
3. Microcontroller Interface: Sends the decoded signals to the robot's microcontroller, which then controls motors, actuators, and other systems accordingly.

5.5 LM2596 DC-DC Buck Converter:



Fig 5.5 -LM2596 DC-DC Converter

LM2596 DC-DC Converter is shown in Fig 5.5, The LM2596 DC-DC buck converter is an essential component in the bomb disposal robot project, providing efficient power regulation to various parts of the robot. It is a versatile and widely used switching regulator that steps down a higher voltage to a lower, stable voltage required by the robot's components such as motors, sensors, and microcontrollers. This ensures the robot operates efficiently without overloading or damaging sensitive electronics.

5.5.1 Function of the LM2596 Buck Converter:

The LM2596 is a step-down voltage regulator, meaning it reduces the input voltage to a desired, lower output voltage. This is particularly important in a robot that uses components with different voltage requirements. For example, the motors may operate at 12V, while the microcontroller and sensors require a 5V supply. The LM2596 converts the higher voltage, such as 12V or 24V from the robot's battery to the lower, stable voltage needed for these components.

The LM2596 DC-DC Buck Converter is a versatile and useful tool for converting a higher input voltage to a lower output voltage. It can be used to power multiple devices with different voltage requirements from a single source. It has built-in over-current protection for added safety.

5.6 Switch :



Fig 5.6 - Switch

switch in Fig 5.6, In the switch is used to control the robot's power and functionality. A power switch turns the robot on and off, ensuring safe operation by preventing accidental activation. Push-button switches control specific actions like moving the robot, activating the gripper arm, or starting the camera. An emergency stop switch is included for immediate shutdown during critical situations, halting the robot's systems to ensure safety. Relay switches may also be used to manage high-power components such as motors. These switches ensure precise control and safety in hazardous bomb disposal operations.

5.6.1 Role of the Switch in the Switch:

The switch is integral to ensuring safe operation of the bomb disposal robot in hazardous environments. Its primary roles include:

1. **Power Control:** The switch ensures the robot can be powered up or down safely. This includes switching the main power supply on or off, as well as powering individual components like sensors, motors, or the communication system.

2. Safety: The switch, especially the emergency stop switch, is critical in bomb disposal missions where quick responses are required. If a problem occurs, such as the robot getting too close to a bomb, the emergency stop immediately halts operations, minimizing risk.
3. Functionality Control: Switches control specific functions such as the movement of the robot, the manipulation of objects with the robotic arm or gripper, and the activation of sensors. This allows the operator to handle the bomb or other objects carefully.

5.7 Jumper Wire:



Fig 5.7 - Jumper Wire

Jumper wires Fig 5.7, are used to establish electrical connections between various components, such as the microcontroller, sensors, motors, and power supply. These flexible wires have connectors at both ends and are used to transfer signals and power between parts of the robot. Jumper wires simplify the prototyping process by providing quick, reliable connections without the need for soldering, allowing easy adjustments and troubleshooting. They are essential for linking components on the robot's control board and connecting subsystems like the motor driver, sensors, and communication modules.

Jumper wires are used to transfer electrical signals and power between different parts of the robot. In the case of a bomb disposal robot, these wires are vital for linking critical systems, such as the microcontroller, motor drivers, sensor arrays, communication modules, and power regulators. For example, jumper wires connect the microcontroller to the motor driver, allowing the robot to control its movement. They also connect sensors like cameras or force sensors to the microcontroller, enabling real-time feedback and precise control.

5.7.1 Advantages in Jumper Wire:

1. **Quick Assembly and Modifications:** Jumper wires allow rapid prototyping, making it easy to assemble and modify the robot's circuits during development. This flexibility is important as designs may evolve during testing.
2. **Ease of Troubleshooting:** They make it simple to disconnect and reconnect parts, which is valuable when troubleshooting electrical connections or replacing faulty components.
3. **Safe and Reliable:** Jumper wires help ensure a secure, reliable connection between components without the need for soldering, minimizing the risk of damage during assembly and adjustments.
4. **Versatility:** These wires can be used for a variety of connections within the robot, from low power control signals to higher-power connections for motors and actuators.

5.8 Wheel Screws :



Fig 5.8 - Wheel screws

wheel screws Fig 5.8, are used to secure the wheels to the robot's chassis, allowing for stable movement and mobility. These screws attach the wheels to the motor shafts or axles, ensuring that they rotate smoothly when powered by the motors. Wheel screws play a critical role in ensuring that the robot can move effectively across different terrains, providing the necessary traction and stability for the robot to navigate safely during bomb disposal operations. Proper fastening of the wheel screws ensures that the wheels remain securely attached, preventing accidents or malfunctions during use.

1. **Stability and Safety:** The wheel screws ensure that the wheels stay securely attached to the robot during movement, preventing dislodging or slipping, especially when manoeuvring through challenging terrains or environments.
2. **Smooth Movement:** Properly installed wheel screws enable the wheels to rotate freely without obstruction, contributing to smooth and precise movement, essential for navigating safely around potential hazards.
3. **Durability:** Since bomb disposal robots often operate in harsh conditions, the durability of wheel screws is important. They need to withstand vibrations, impacts, and environmental stressors without loosening or breaking.

5.9 Nuts & Nuts:



Fig 5.9 - Nut & Nut

Nut Fig 5.9, are used in conjunction with screws to secure various components, including the wheels, motors, and structural parts, to the robot's chassis. The nuts fit onto the threaded ends of screws, creating a strong and stable connection that holds the components in place. This ensures that parts like the wheels, motor mounts, and other critical systems remain securely attached during operation. Nuts are essential

for maintaining the robot's structural integrity, allowing it to function smoothly and safely in hazardous bomb disposal environments.

5.10 DC Motor :-

A **DC motor** (Direct Current motor) is an electromechanical device that converts direct current electrical energy into mechanical energy. It is a vital component in robotics, automation systems, and various industrial applications due to its simplicity, efficiency, and ease of control.

Key Features of DC Motors: -

- **Simple Design:** Comprises essential components like a stator, rotor (armature), commutator, and brushes.
- **Speed Control:** Achieved by varying the supply voltage or current, making it suitable for precision applications.
- **High Starting Torque:** Ideal for applications requiring significant torque at low speeds.
- **Variety:** Available in brushed and brushless types, catering to diverse operational needs.

Types of DC Motors:

- **Brushed DC Motor:** Uses brushes and a commutator for current transfer to the rotor. Suitable for basic, low-cost applications.
- **Brushless DC Motor (BLDC):** Eliminates brushes, ensuring longer life, higher efficiency, and reduced maintenance.
- **DC Motor:** Includes a gearbox to reduce speed while increasing torque.

Applications:

- Robotics for locomotion and joint movements.
- Conveyor belts and automated systems.
- Motorized toys and vehicles.
- Industrial tools like drills and screwdrivers.

Advantages:

- Smooth operation and reliable performance.
- Precise control over speed and position.
- Compact size with lightweight designs for portable devices.

Relevance to My Project:

In this project, the DC motor is utilized for enabling directional movement in the robotic system. The motor is controlled via SPI communication Protocol ensuring robust and efficient data transmission. The high torque and speed controllability of the DC motor make it an ideal choice for achieving precise navigation and maneuverability in robotic operations.

The integration of DC motors underscores the project's focus on efficiency and practical implementation of electromechanical systems.

5. Applications of project: -

1. It is used for spying and diffusing the bomb.
2. It can be used by: police: in hostage situations

3. Military: For reconnaissance missions.
4. Fire: To provide video feedback of the site for analysis.
5. It can be used to lift light weight medicine packages and other necessary goods in emergency cases

7. Program:-

7.1 . Receiver Side Code : -

```
#include <RH_ASK.h>

#include <ServoTimer2.h>

#include <SPI.h>

ServoTimer2 mg995Servo1;

ServoTimer2 mg995Servo2;

RH_ASK rf_driver;

const int servoPin1 = 8;

const int servoPin2 = 10;

void setup() {

    mg995Servo1.attach(servoPin1);

    mg995Servo2.attach(servoPin2);

    mg995Servo1.write(20);

    mg995Servo2.write(20);

    delay(1000);

    digitalWrite(2, LOW);

    digitalWrite(3, LOW);

    digitalWrite(4, LOW);

    digitalWrite(5, LOW);

    Serial.begin(9600);

    if (!rf_driver.init()) {

        Serial.println("RF module failed to initialize");

        while (1); // Halt the program if RF initialization fails

    }

}
```

```

pinMode(2, OUTPUT);

pinMode(3, OUTPUT);

pinMode(4, OUTPUT);

pinMode(5, OUTPUT);

}

void loop() {

  uint8_t buf[RH_ASK_MAX_MESSAGE_LEN];

  uint8_t buflen = sizeof(buf);

  if (rf_driver.recv(buf, &buflen)) {

    buf[buflen] = '\0';

    String receivedMessage = String((char*)buf);

    Serial.print("Message Received: ");

    Serial.println(receivedMessage);

    if (receivedMessage == "motor_forword") {

      digitalWrite(2, HIGH);

      digitalWrite(3, LOW);

      digitalWrite(4, LOW);

      digitalWrite(5, HIGH);

      delay(500);

    } else if (receivedMessage == "motor_backword") {

      digitalWrite(2, LOW);

      digitalWrite(3, HIGH);

      digitalWrite(4, HIGH);

      digitalWrite(5, LOW);

      delay(500);

    } else if (receivedMessage == "motor_right") {

      digitalWrite(2, LOW);

      digitalWrite(3, HIGH);

      digitalWrite(4, LOW);


```

```

    digitalWrite(5, HIGH);

    delay(500);

} else if (receivedMessage == "motor_left") {

    digitalWrite(2, HIGH);

    digitalWrite(3, LOW);

    digitalWrite(4, HIGH);

    digitalWrite(5, LOW);

    delay(500);

} else if (receivedMessage == "servo_1") {

    for (int angle = 0; angle <= 180; angle++) {

        mg995Servo1.write(angle);

        delay(15);

    }

    delay(1000);

    for (int angle = 180; angle >= 0; angle--) {

        mg995Servo1.write(angle);

        delay(15);

    }

} else if (receivedMessage == "servo_2") {

    for (int angle = 0; angle <= 180; angle++) {

        mg995Servo2.write(angle);

        delay(15);

    }

    delay(1000);

    for (int angle = 180; angle >= 0; angle--) {

        mg995Servo2.write(angle);

        delay(15);

    }

    digitalWrite(2, LOW);

    digitalWrite(3, LOW);

    digitalWrite(4, LOW);

    digitalWrite(5, LOW);

```

```

    }
}
}

```

7.2 . Transmitter Code

```

#include <RH_ASK.h>

#include <SPI.h> // Not used but required for compatibility
RH_ASK rf_driver;

void setup() {
    Serial.begin(9600); // Initialize serial communication
    if (!rf_driver.init()) {
        Serial.println("RF module failed to initialize");
    }

    pinMode(2, INPUT); // Button 1
    pinMode(3, INPUT); // Button 2
    pinMode(4, INPUT); // Button 3
    pinMode(5, INPUT); // Button 4
    pinMode(6, INPUT); // Button 5
    pinMode(7, INPUT); // Button 6

}

void loop() {
    if (digitalRead(2) == HIGH) {
        Serial.println("Message Transmitted: 1 ");
        const char *msg = "motor_forword";
        rf_driver.send((uint8_t *)msg, strlen(msg));
        rf_driver.waitPacketSent();
    }

    else if (digitalRead(3) == HIGH) {

```



```

Serial.println("Message Transmitted: 2");

const char *msg = "motor_backword";

rf_driver.send((uint8_t *)msg, strlen(msg));

rf_driver.waitPacketSent();

}

else if (digitalRead(4) == HIGH) {

Serial.println("Message Transmitted: 3 ");

const char *msg = "motor_right";

rf_driver.send((uint8_t *)msg, strlen(msg));

rf_driver.waitPacketSent();

}

else if (digitalRead(5) == HIGH) {

Serial.println("Message Transmitted: 4 ");

const char *msg = "motor_left";

rf_driver.send((uint8_t *)msg, strlen(msg));

rf_driver.waitPacketSent();

}

else if (digitalRead(6) == HIGH) {

Serial.println("Message Transmitted: 5 ");

const char *msg = "SERVO_1";

rf_driver.send((uint8_t *)msg, strlen(msg));

rf_driver.waitPacketSent();

}

else if (digitalRead(7) == HIGH) {

Serial.println("Message Transmitted: 6 ");

const char *msg = "SERVO_2";

rf_driver.send((uint8_t *)msg, strlen(msg));

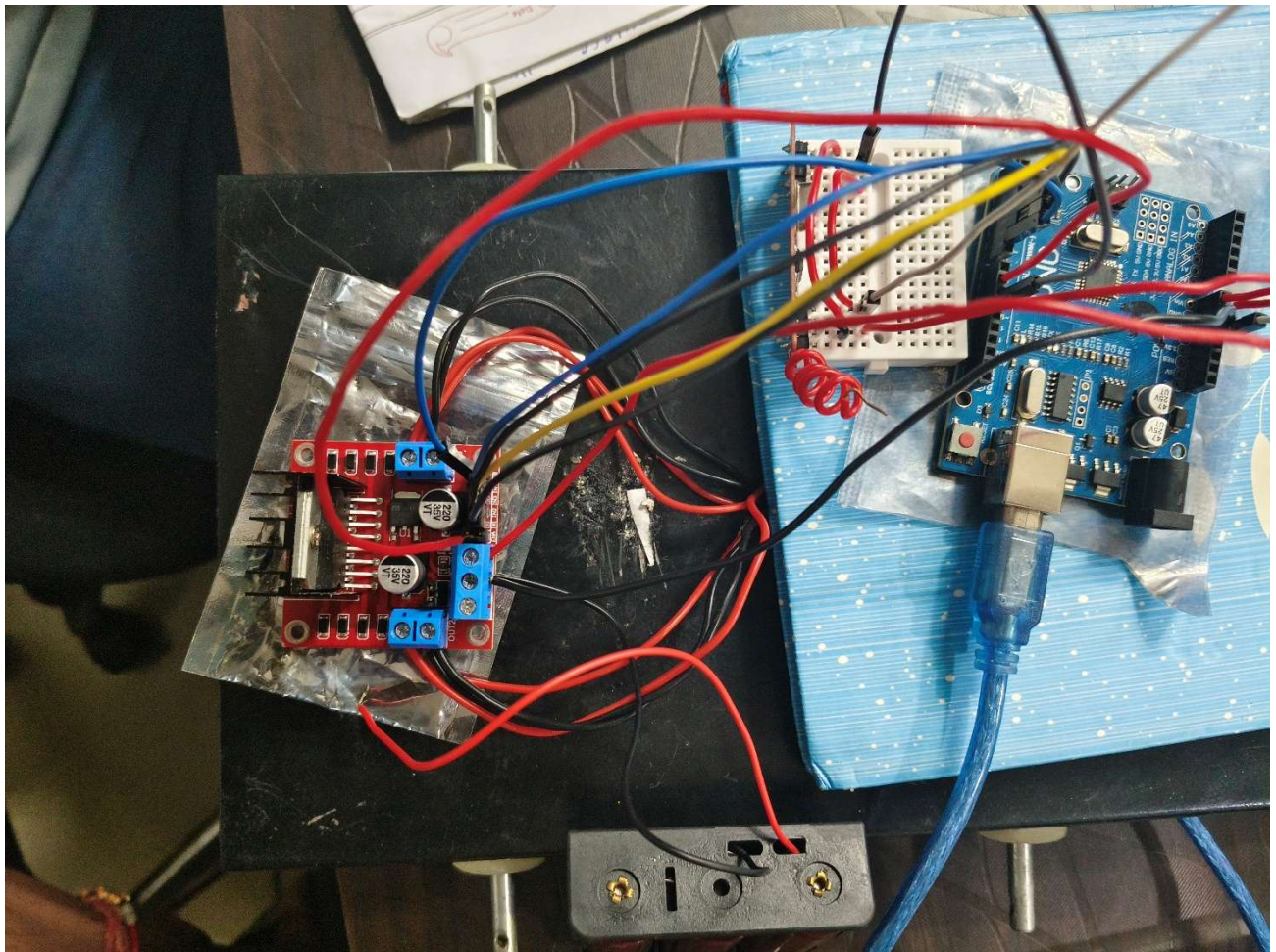
rf_driver.waitPacketSent();

}

}

```

8. Project Testing: -



9. Components Prize: -

S.No.	Component Name	Prize
1.	2 Arduino Uno	1100
2.	L298N Motor Driver	399
3.	Gripper (Robotic Arm)	350
4.	RF Controller Tx Rx	250
5.	LM2596 DC – DC Buck Converter	300
6.	Switches	400
7.	Jumper Wires	60
8.	Wheel Screws	480
9.	Nut & Nut	100
10.	Motors	1200
11.	Robot Body case	300

12.	Battery (lithium cell)	300
13.	2 Servo Motor	700
14.		
15.		

10. Conclusion

In the bomb disposal robot project integrates various components, including motors, sensors, actuators, and power management systems, to create a functional, safe, and efficient robotic system. Key elements like the L298N motor driver, LM2596 DC-DC converter, and jumper wires ensure reliable performance, while switches, wheel screws, and nuts provide essential control, stability, and security. The robot is designed to navigate hazardous environments, allowing operators to handle dangerous situations remotely, minimizing risk. Overall, the project demonstrates how advanced robotics can be applied in real-world safety-critical applications like bomb disposal.

12. Reference:-

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